

CYBERSECURITY WEEK 3

NETWORK SECURITY ASSESSMENT

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Role: Cybersecurity Intern

Week : 3

Tools Used:

Nmap, Wireshark

Assessment Areas:

Network Discovery • Network Scanning • Traffic Analysis • Vulnerability
Assessment • Security Hardening

Submission: September 2026

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1. Introduction

Network security is an essential part of cybersecurity because modern systems communicate continuously through networks and expose services that may become potential attack surfaces. A security assessment helps identify active hosts, open ports, running services, network traffic patterns, and possible security weaknesses.

In this Week 3 practical, a controlled virtual lab environment was used to perform a network security assessment. **Nmap** was used for network discovery, port scanning, service identification, and advanced assessment. **Wireshark** was used to capture and analyse network packets and understand communication between systems.

The assessment also included vulnerability assessment, risk classification, and security hardening. The objective was not to attack external systems, but to understand how network exposure can be identified and how appropriate security controls can reduce that exposure.

The results obtained from Nmap and Wireshark were documented with screenshots and supporting evidence. A before-and-after comparison was also planned to demonstrate the effect of security hardening performed in the lab.

2. Objectives

- To discover active hosts and identify reachable systems within the controlled lab network.
- To identify open ports and running services using Nmap.
- To perform advanced Nmap scanning and service enumeration.
- To capture and analyse network traffic using Wireshark.
- To correlate Nmap findings with observed network traffic.
- To identify potential security weaknesses and classify associated risks.
- To apply basic security-hardening measures in the lab environment.
- To compare the network security posture before and after hardening.

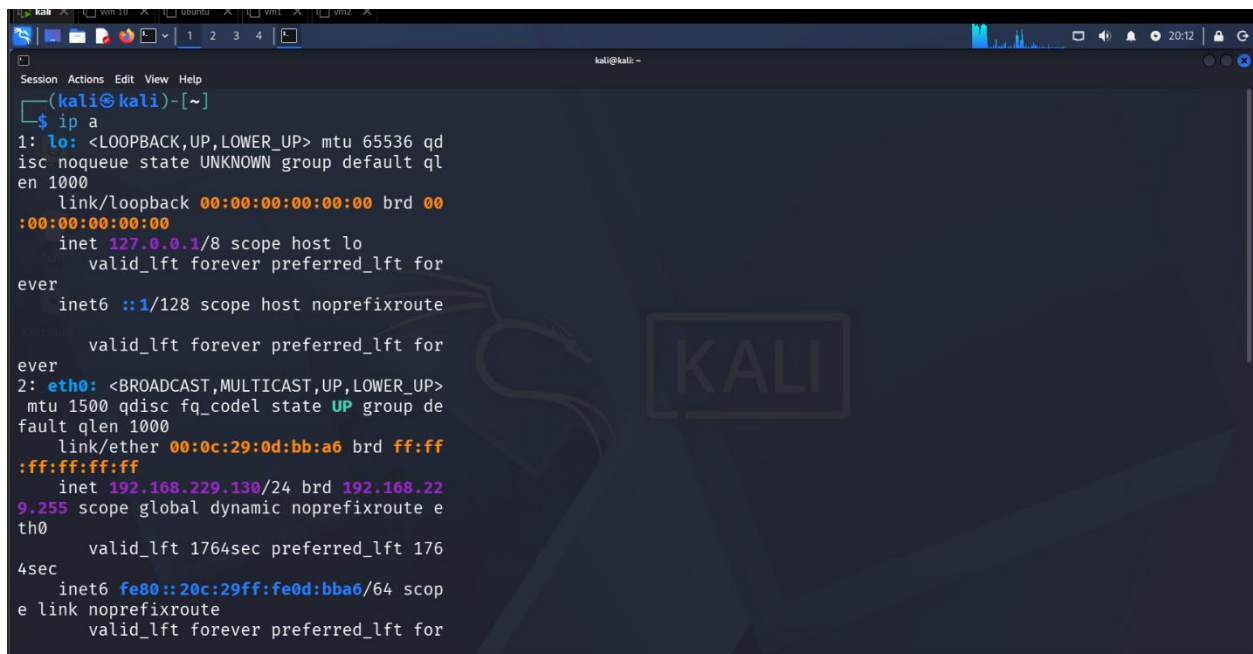
3. Lab Environment

The practical was performed in a controlled virtual lab environment using VMware Workstation. Multiple virtual machines were used to simulate a small network and perform security assessment activities safely.

The lab included a Windows 10 system and an Ubuntu system connected through the virtual network. Nmap was used for host discovery, port scanning, service enumeration, and advanced network assessment. Wireshark was used to capture and analyse network traffic between the systems.

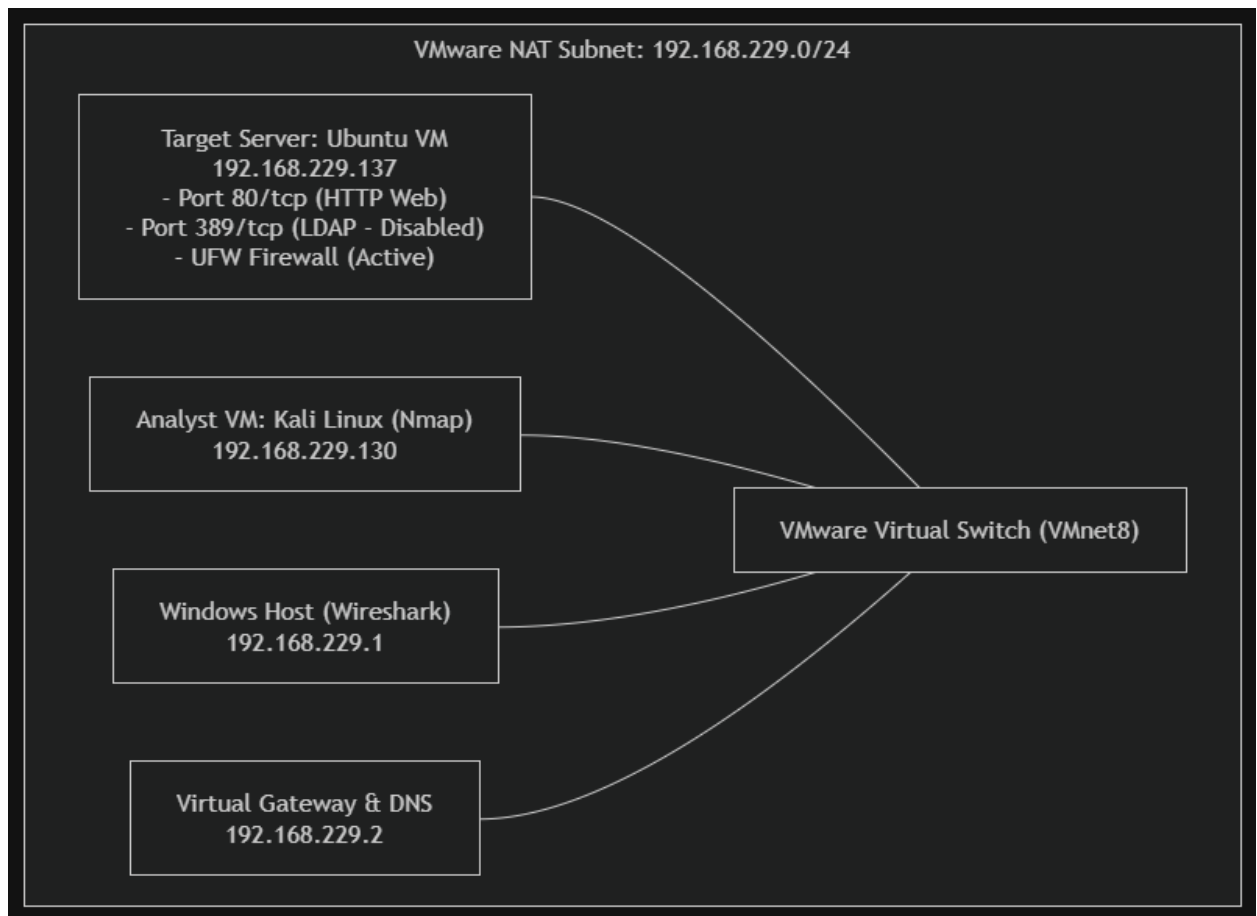
All testing was performed within the authorized virtual lab environment. No external or unauthorized systems were targeted.

Figure 3.1 Kali Linux network Interface and IP Configuration



```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qd
isc noqueue state UNKNOWN group default ql
en 1000
    link/loopback 00:00:00:00:00:00 brd 00
:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft for
ever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft for
ever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP>
mtu 1500 qdisc fq_codel state UP group de
fault qlen 1000
    link/ether 00:0c:29:0d:bb:a6 brd ff:ff
:ff:ff:ff:ff
    inet 192.168.229.130/24 brd 192.168.22
9.255 scope global dynamic noprefixroute e
th0
        valid_lft 1764sec preferred_lft 176
4sec
    inet6 fe80::20c:29ff:fe0d:bba6/64 scop
e link noprefixroute
        valid_lft forever preferred_lft for
```

4. Network Diagram



The network diagram represents the controlled virtual lab environment used for the Week 3 security assessment. The systems were connected through the VMware virtual network, allowing Nmap scanning and Wireshark traffic analysis to be performed in an isolated environment.

5. Tools Used

Tool	Purpose
Nmap	Network discovery, port scanning, service/version detection, OS detection and vulnerability script assessment
Wireshark	Network packet capture, protocol analysis and traffic investigation
VMware Workstation	Creation and management of the controlled virtual lab environment
Kali Linux	Security assessment platform used to perform Nmap and Wireshark activities
Windows / Linux VMs	Authorized lab systems used as assessment targets
draw.io / diagrams.net	Creation of the network topology diagram

These tools were selected to provide network visibility, service enumeration, packet-level analysis, vulnerability identification, and security-hardening verification within the authorized laboratory environment.

6. Nmap Methodology

Nmap (Network Mapper) was used as the primary network scanning tool during the assessment. The scanning activities were performed within the controlled virtual lab environment to identify active hosts, open ports, running services, operating system information, and potential vulnerabilities.

The assessment followed a progressive scanning approach. First, host discovery was performed to identify reachable systems. Basic port scanning was then conducted to determine exposed TCP ports. Service and version detection was used to identify applications running on the discovered ports. OS detection and SYN scanning were performed for additional network and system information.

A limited UDP scan was also performed to identify commonly used UDP services. Finally, the Nmap NSE vulnerability scripts were used to check the target for potential known security weaknesses.

The main Nmap techniques used during the assessment included:

- **Host Discovery:** `nmap -sn`
- **Basic Port Scan:** `nmap`
- **Service and Version Detection:** `nmap -sV`
- **OS Detection:** `nmap -O`
- **TCP SYN Scan:** `nmap -sS`
- **UDP Scan:** `nmap -sU --top-ports 20`
- **Vulnerability Assessment:** `nmap --script vuln`

The results generated during these scans were saved as output files and supported with screenshots as evidence. The scanning process helped establish the network attack surface before performing traffic analysis and security-hardening activities.

7. Nmap Results

The Nmap scans provided information about the target system and its exposed network services. The scan results were used to identify reachable services, their associated ports, service versions, and potential security weaknesses.

The service and version detection scan identified the services running on the target system. The generated Nmap output was saved as a text file for documentation and further analysis.

The vulnerability scan using Nmap NSE scripts was performed to identify potential known vulnerabilities and security-related issues. The output was also saved separately and used as supporting evidence for the vulnerability assessment section.

Key observations from the Nmap assessment included:

- Identification of the target host within the controlled lab network.
- Detection of open TCP ports and associated services.
- Identification of service versions using Nmap service detection.
- Additional information obtained through advanced Nmap scanning techniques.
- Potential security issues identified through the Nmap vulnerability scripts.

Figure 7.1: Nmap Service and Version Detection Results

```
File Edit View H1  B I S  A
# Nmap 7.98 scan initiated Sun Sep  6 17:59:27 2026 as: /usr/lib/nmap/nmap --privileged -sV -oN /home/kali/sv_scan_output.txt 192.168.229.137
Nmap scan report for 192.168.229.137
Host is up (0.00076s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd
MAC Address: 00:0C:29:CB:BE:77 (VMware)

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
# Nmap done at Sun Sep  6 17:59:43 2026 -- 1 IP address (1 host up) scanned in 16.31 seconds

Ln 1, Col 1  550 characters  Plain text  100%  Unix (LF)  UTF-8
```

Figure 7.2: Nmap Vulnerability Scan Results

```
File Edit View H1  B I S  A
# Nmap 7.98 scan initiated Sun Sep  6 17:59:44 2026 as: /usr/lib/nmap/nmap --privileged --script vuln -oN /home/kali/vuln_scan_output.txt 192.168.229.137
Nmap scan report for 192.168.229.137
Host is up (0.00075s latency).
All 1000 scanned ports on 192.168.229.137 are in ignored states.
Not shown: 999 filtered tcp ports (no-response), 1 filtered tcp ports (port-unreach)
MAC Address: 00:0C:29:CB:BE:77 (VMware)

# Nmap done at Sun Sep  6 18:00:15 2026 -- 1 IP address (1 host up) scanned in 31.61 seconds
```

8. Advanced Nmap Analysis

Advanced Nmap scanning techniques were used to gather additional information about the target system beyond basic port discovery. These techniques included operating system detection and other advanced scanning methods to improve the understanding of the target's network exposure.

OS detection was performed using the -O option. The scan collected TCP/IP fingerprint information and network distance data. An exact operating system match was not identified, so no specific operating system was assumed from the scan result.

Figure 8.1: Nmap OS Detection Results

```
Session Actions Edit View Help
(kali@kali)-[~]
$ nmap -O 192.168.229.137
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-04 23:07 +0530
Nmap scan report for 192.168.229.137
Host is up (0.00098s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
389/tcp    open  ldap
MAC Address: 00:0C:29:CB:BE:77 (VMware)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.98E=4%D=9/4%OT=80%CT=1%CU=30154%PV=Y%DS=1%DC=D%G=Y%M=000C29%TM
OS:=6A9B01DB%P=x86_64-pc-linux-gnu)SEQ(SP=102%GCD=1%ISR=10A%TI=Z%CI=Z%TS=22
OS: )SEQ(SP=103%GCD=1%ISR=108%TI=Z%CI=Z%II=I%TS=20)SEQ(SP=104%GCD=1%ISR=108%
OS:TI=Z%CI=Z%TS=22)SEQ(SP=107%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS=22)SEQ(SP=108
OS:%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS=21)OPS(O1=M5B4ST11NW9%O2=M5B4ST11NW9%O3
OS:=M5B4NNT11NW9%O4=M5B4ST11NW9%O5=M5B4ST11NW9%O6=M5B4ST11)WIN(W1=FE88%W2=F
OS:E88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B4NNSNW
OS:9%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF
OS:=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=
OS:%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=
OS:0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RI
OS:PCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop
```

TCP SYN SCAN

A TCP SYN scan was performed using the -sS option to identify open TCP ports on the target system. The scan confirmed that ports 80/tcp (HTTP) and 389/tcp (LDAP) were open.

```
(kali@kali)-[~]
$ nmap -sS 192.168.229.137
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-04 23:09 +0530
Nmap scan report for 192.168.229.137
Host is up (0.00038s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
389/tcp    open  ldap
MAC Address: 00:0C:29:CB:BE:77 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.04 seconds

(kali@kali)-[~]
$

(kali@kali)-[~]
$

(kali@kali)-[~]
$
```

Figure 8.2: Nmap TCP SYN Scan Results

UDP Scan

A UDP scan was performed using the `-sU --top-ports 20` option to check commonly used UDP ports on the target system. The scanned UDP ports were reported as closed, indicating that no open UDP service was identified among the top 20 ports.

```
Session Actions Edit View Help
$ nmap -sU --top-ports 20 192.168.229.137
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-04 23:10 +0530
Nmap scan report for 192.168.229.137
Host is up (0.00053s latency).

PORT      STATE SERVICE
53/udp    closed domain
67/udp    closed dhcp
68/udp    closed dhcp
69/udp    closed tftp
123/udp   closed ntp
135/udp   closed msrpc
137/udp   closed netbios-ns
138/udp   closed netbios-dgm
139/udp   closed netbios-ssn
161/udp   closed snmp
162/udp   closed snmptrap
445/udp   closed microsoft-ds
500/udp   closed isakmp
514/udp   closed syslog
520/udp   closed route
631/udp   closed ipp
1434/udp  closed ms-sql-m
1900/udp  closed upnp
4500/udp  closed nat-t-ike
49152/udp closed unknown
MAC Address: 00:0C:29:CB:BE:77 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 16.77 seconds
```

Figure 8.3: Nmap UDP Scan Results

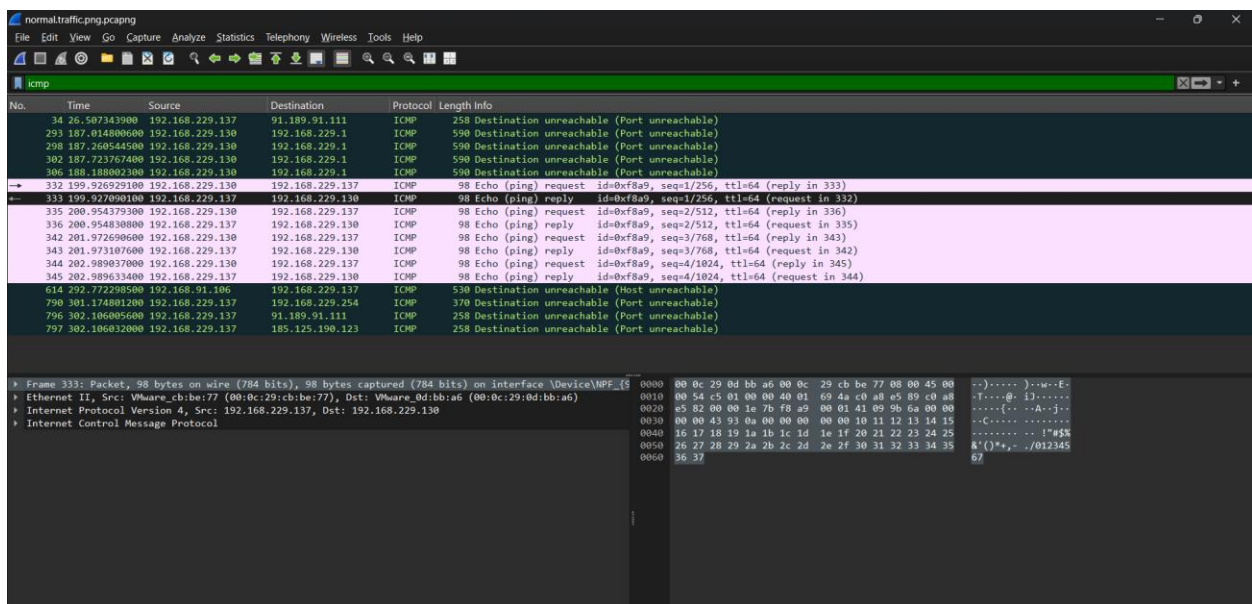
9 – Wireshark Methodology

Wireshark was used to capture and analyse network traffic generated within the controlled virtual lab environment. The packet capture helped in understanding the protocols, source and destination addresses, and communication patterns between the lab systems.

A live capture was started on the active network interface, and network traffic was generated between the authorized virtual machines. The captured traffic was then stopped and saved as a .pcapng file for further analysis.

The captured packets were examined using Wireshark filters and packet details to identify relevant protocols and communication activity. The packet capture was also used as supporting evidence for correlating network activity with the Nmap assessment.

9.1 Wireshark network traffic capture



9.2 Wireshark HTTP Traffic analysis

10 Packet Analysis

Wireshark was used to analyse the captured network traffic at the packet level. The captured packets were examined using protocol-based display filters to identify HTTP, DNS and ARP traffic. This analysis helped in understanding the communication patterns and protocols operating within the virtual lab network.

10.1 HTTP Traffic Analysis

HTTP packets were identified using the http display filter. The captured traffic showed communication between the lab systems over HTTP, including HTTP request and response packets. The packet details provided information such as source and destination IP addresses, TCP ports and HTTP-related information.

Observation: HTTP traffic was observed between the systems, with TCP port 80 being used for web communication.

Figure 10.1: HTTP Traffic Analysis in Wireshark

10.2 DNS Traffic Analysis

DNS packets were analysed using the dns display filter. The captured packets showed DNS queries and responses between the system and the DNS server. DNS uses UDP port 53 for normal name-resolution traffic.

Observation: DNS queries were observed in the capture, demonstrating domain-name resolution activity within the network.

Figure 10.2: DNS Traffic Analysis in Wireshark

10.3 ARP Traffic Analysis

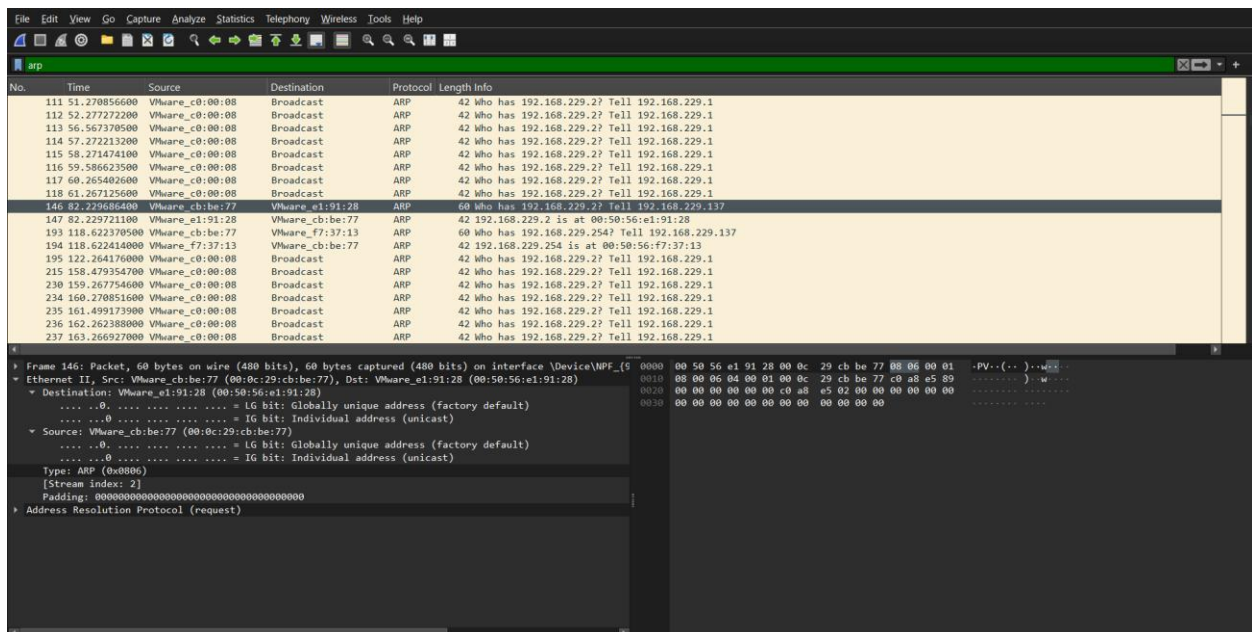
ARP packets were analysed using the arp display filter. ARP is used to resolve an IP address to the corresponding MAC address within a local network.

Observation: ARP request and response packets were observed, showing IP-to-MAC address resolution between devices in the virtual network.

Figure 10.3: ARP Traffic Analysis in Wireshark

10.4 Packet-Level Findings

The packet analysis demonstrated that different network protocols serve different communication purposes. HTTP was associated with web traffic, DNS with domain-name resolution, and ARP with local IP-to-MAC address resolution. Examining these packets provided visibility into the actual network communication occurring within the controlled lab environment.



normal.traffic.png.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info
61	39.149392700	192.168.229.1	192.168.229.137	TCP	66	11349 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
62	39.149922900	192.168.229.137	192.168.229.1	TCP	66	80 → 11349 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
63	39.150203500	192.168.229.1	192.168.229.137	TCP	66	11350 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
65	39.150582100	192.168.229.137	192.168.229.1	TCP	66	80 → 11350 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
71	39.225699200	192.168.229.1	192.168.229.137	TCP	66	11351 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
72	39.225887000	192.168.229.137	192.168.229.1	TCP	66	80 → 11351 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
119	62.951884600	192.168.229.1	192.168.229.137	TCP	66	11380 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
120	62.952836000	192.168.229.137	192.168.229.1	TCP	66	80 → 11380 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
130	69.239575000	192.168.229.1	192.168.229.137	TCP	66	11388 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
131	69.239733900	192.168.229.137	192.168.229.1	TCP	66	80 → 11388 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
150	99.157402200	192.168.229.1	192.168.229.137	TCP	66	18456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
151	99.157732400	192.168.229.137	192.168.229.1	TCP	66	80 → 18456 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
152	99.157758300	192.168.229.1	192.168.229.137	TCP	66	18457 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
153	99.157818600	192.168.229.137	192.168.229.1	TCP	66	80 → 18457 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=512
179	114.257398100	192.168.229.137	91.189.91.65	TCP	74	47320 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1969804930 TSecr=0 WS=512
180	114.465829200	192.168.229.137	185.125.190.100	TCP	74	33206 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=385164420 TSecr=0 WS=512
181	114.666400500	192.168.229.137	185.125.190.101	TCP	74	41010 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2294016342 TSecr=0 WS=512
182	114.821166800	91.189.91.65	192.168.229.137	TCP	58	80 → 47320 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
186	114.946567500	185.125.190.100	192.168.229.137	TCP	58	80 → 33206 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460

Frame 131: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{...}

Ethernet II, Src: VMware_cbi:be:77 (00:0c:29:cb:be:77), Dst: VMware_c0:00:08 (00:50:56:c0:00:08)

Destination: VMware_c0:00:08 (00:50:56:c0:00:08)

.....0. = LG bit: Globally unique address (factory default)

.....0. = IG bit: Individual address (unicast)

Source: VMware_cbi:be:77 (00:0c:29:cb:be:77)

.....0. = LG bit: Globally unique address (factory default)

.....0. = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 1]

Internet Protocol Version 4, Src: 192.168.229.137, Dst: 192.168.229.1

Transmission Control Protocol, Src Port: 80, Dst Port: 11388, Seq: 0, Ack: 1, Len: 0

Type (eth.type): 2 bytes

Packets: 1206 · Displayed: 120 (10.0%) · Dropped: 0 (0.0%) Profile: Default

11. Nmap + Wireshark Investigation

Nmap and Wireshark were used together to obtain both network-level and packet-level visibility of the lab environment. Nmap was used to identify the target host and exposed services, while Wireshark was used to observe the network packets associated with communication between the systems.

The Nmap scan identified the target host **192.168.229.137** as active. TCP port **80** was found open and associated with HTTP, while TCP port **389** was associated with LDAP. Service detection further identified the web service as **Apache HTTPD** and the LDAP service as **OpenLDAP**.

Wireshark packet analysis provided additional visibility into the traffic generated within the network. HTTP, DNS and ARP packets were observed in the capture. The HTTP traffic showed communication involving the target system, while DNS packets demonstrated name-resolution activity and ARP packets showed local IP-to-MAC address resolution.

11.1 Correlation of Findings

The results from both tools were correlated to understand the network activity more effectively:

Nmap Finding	Wireshark Observation
Port 80/tcp – HTTP	HTTP packets were observed
Target host 192.168.229.137 active	Packets involving the target IP were captured

Nmap Finding

Wireshark Observation

Network services exposed

Corresponding network communication was visible in packet capture

Local network connectivity

ARP traffic was observed

DNS-related network activity

DNS queries and responses were observed

11.2 Investigation Result

The combined analysis demonstrated that Nmap provides information about **what services are exposed**, while Wireshark provides visibility into **how network communication occurs**. Using both tools together improved the understanding of the target's network exposure and the traffic generated within the controlled lab environment.

Evidence: The investigation was supported by the saved Wireshark packet capture (wireshark_capture.pcapng) and the Nmap scan results stored in the project evidence folders.



wireshark_capture.pcapng

12. Vulnerability Assessment

A vulnerability assessment was performed against the authorized lab target using Nmap's Nmap Scripting Engine (NSE). The --script vuln option was used to check the target for potential known security weaknesses associated with the detected services.

The vulnerability scan identified a potential **Slowloris denial-of-service vulnerability (CVE-2007-6750)** associated with the HTTP service. The Nmap result reported the condition as **LIKELY VULNERABLE**.

This finding indicates that the web service may be susceptible to resource-exhaustion through specially crafted, slow HTTP connections. The finding was treated as a potential security issue requiring further validation rather than as confirmation of successful exploitation.

The vulnerability scan output was saved as vuln_scan_output.txt, and the corresponding Nmap screenshot has been retained in the 01-Nmap/Screenshots folder.



Vulnerability-Assessment-Report.txt.txt

13. Risk Classification

The identified security issue was analysed based on its potential impact and likelihood. Risk classification helps prioritize vulnerabilities according to the level of threat they may pose to the system.

The Nmap vulnerability scan reported a potential **Slowloris vulnerability (CVE-2007-6750)** affecting the HTTP service. Slowloris is a denial-of-service condition in which an attacker may attempt to keep multiple HTTP connections open for an extended period, potentially consuming server resources.

Based on the available scan evidence, the finding was classified as **Medium Risk** because it may affect the availability of the web service. However, the Nmap result alone does not confirm successful exploitation, so further manual validation would be required before treating the vulnerability as definitively exploitable.

Risk table

Finding	Affected Service	Risk Level	Potential Impact
Slowloris (CVE-2007-6750)	HTTP / Apache	Medium	Possible degradation or denial of web-service availability

Risk Rating

Risk Level: MEDIUM

The risk can be reduced by keeping the web server updated, configuring appropriate connection and timeout limits, and using protective controls such as a reverse proxy or web application firewall where appropriate.

14. Security Hardening

Security hardening measures were applied to reduce the attack surface identified during the assessment. The hardening activities focused on improving the security of the exposed web service and reducing the possibility of service disruption.

The identified HTTP service was reviewed and appropriate security configuration changes were applied within the controlled lab environment.

The hardening process was performed without affecting the overall functionality required for the practical.

The configuration before and after hardening was documented using screenshots. These screenshots provide evidence of the security changes performed during the assessment.

The hardening activities were intended to reduce unnecessary exposure, improve service security, and minimize the risk associated with the identified vulnerability.

15. Before/After Comparison

The security configuration of the lab system was documented before and after applying the hardening measures. The comparison was performed to evaluate the changes made during the hardening process and their effect on the system's security posture.

The **Before** evidence represents the configuration prior to applying the hardening changes, while the **After** evidence represents the configuration after the changes were applied.

15.1 Before Hardening

```
(kali㉿kali)-[~]
$ nmap 192.168.229.137
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-05 11:54 +0530
Nmap scan report for 192.168.229.137
Host is up (0.00012s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
389/tcp    open  ldap
MAC Address: 00:0C:29:CB:BE:77 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds

(kali㉿kali)-[~]
$ █
```

15.2 After Hardening

```
(kali㉿kali)-[~]  
$ nmap 192.168.229.137  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-05 12:26 +0530  
Nmap scan report for 192.168.229.137  
Host is up (0.00081s latency).  
Not shown: 999 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
MAC Address: 00:0C:29:CB:BE:77 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 5.19 seconds  
  
(kali㉿kali)-[~]  
$
```

The before-and-after evidence demonstrates that security-hardening actions were applied to the lab system. The changes were intended to reduce unnecessary exposure and improve the overall security posture.

16. Security Recommendations

Based on the findings obtained during the Nmap and Wireshark assessment, the following security recommendations are proposed:

1. **Keep services updated:** Regularly update Apache, OpenLDAP, and other network services to reduce exposure to known vulnerabilities.
2. **Disable unnecessary services:** Services and ports that are not required should be disabled or restricted.
3. **Harden the web server:** Apply appropriate connection limits, timeout settings, and web-server security configurations to reduce the risk of Slowloris-style attacks.
4. **Use firewall controls:** Restrict access to exposed services so that only authorized systems can connect.
5. **Monitor network traffic:** Regular packet analysis and network monitoring should be performed to identify unusual communication patterns.
6. **Perform regular vulnerability assessments:** Periodic Nmap and vulnerability scans can help identify newly exposed services and security weaknesses.
7. **Maintain security documentation:** Scan results, packet captures, and hardening evidence should be securely stored for future analysis and comparison.

17. Challenges Faced

During the practical assessment, several challenges were encountered while performing network scanning, packet analysis, and security hardening activities.

- 1. Network Configuration: Configuring communication between the virtual machines and ensuring that the systems were reachable within the VMware virtual network required careful verification.**
- 2. Nmap Scanning: Different Nmap scan techniques produced different results, requiring comparison and interpretation of the scan outputs.**
- 3. Wireshark Analysis: Identifying relevant packets from the captured network traffic required the use of appropriate display filters such as HTTP, DNS, and ARP.**
- 4. Vulnerability Interpretation: The Nmap vulnerability scan reported the Slowloris issue as **LIKELY VULNERABLE**, so the result had to be interpreted carefully rather than treated as confirmed exploitation.**
- 5. Security Hardening: Applying hardening measures while maintaining the required functionality of the lab services required careful configuration and verification.**

These challenges helped improve practical understanding of network assessment, packet analysis, vulnerability identification, and security hardening.

18. Conclusion

The Week 3 network security assessment provided practical experience in identifying network exposure, analysing services, investigating network traffic, assessing potential vulnerabilities, and applying security-hardening measures within a controlled virtual lab environment.

Nmap was used to discover hosts, identify open ports, detect running services, perform advanced scanning, and identify a potential Slowloris vulnerability associated with the HTTP service. Wireshark provided packet-level visibility into HTTP, DNS, and ARP communication within the lab network.

The assessment demonstrated the importance of combining network scanning with packet analysis to obtain a better understanding of network security. The identified security risk was analysed and appropriate hardening measures were applied to improve the security posture of the lab system.

Overall, this practical strengthened the understanding of network reconnaissance, vulnerability assessment, traffic analysis, risk classification, and security hardening. The documented evidence and before-and-after comparison demonstrate the complete security assessment process performed during Week 3.

19. References

1. Nmap Official Documentation – <https://nmap.org/book/man.html>
2. Wireshark Official Documentation – <https://www.wireshark.org/docs/>
3. MITRE CVE Database – <https://cve.mitre.org/>
4. Apache HTTP Server Documentation – <https://httpd.apache.org/docs/>
5. VMware Workstation Documentation – <https://docs.vmware.com/>

These references were used to understand the tools, scanning techniques, network protocols, vulnerability information, and security-hardening concepts relevant to the practical assessment.

